

Functional Central Limit Theorems for Dynamic Point Processes

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(Joint work w. Omer Bobrowski^{2,3} and Robert J. Adler³)

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Penn/Temple Probability Seminar

- ▶ O, Omer Bobrowski, and Robert J. Adler (2023b).
“Functional Limit Theorems for Local Functionals of
Dynamic Point Processes”. *arXiv:2310.17775*

Motivation

- ▶ Random Geometric Graphs



<http://www.cis.upenn.edu/~jshen/vis/GeoNetwork.jpg>

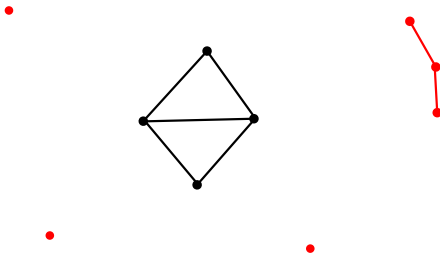
- ▶ M. Haenggi, J. G. Andrews, F. Baccelli, O. Dousse, and M. Franceschetti (2009). “Stochastic geometry and random graphs for the analysis and design of wireless networks”. *IEEE journal on selected areas in communications*
- ▶ Mathew Penrose (2003). *Random geometric graphs*. Oxford Studies in Probability. Oxford University Press

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- ▶ Dynamic (time-changing) setups?

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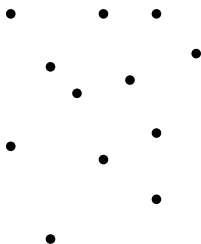


Spatial Birth Death Processes

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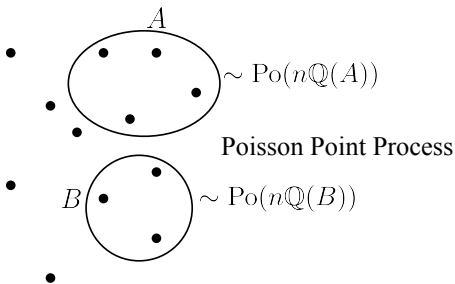


Poisson Point Process

$t = 0$

Spatial Birth Death Processes

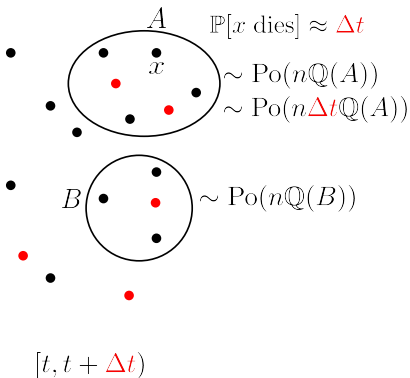
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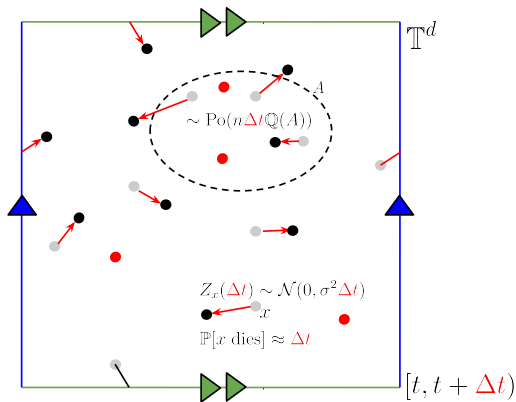
Spatial Birth Death Processes

$\mathbb{Q}$: a probability measure.



Mobility

$\mathbb{Q}$: Haar measure on $\mathbb{T}^d$.



Dynamic Boolean Model

- ▶ J. van den Berg, Ronald Meester, and Damien G. White (1997). “Dynamic Boolean models”. *Stochastic Process. Appl.*
- ▶ Yuval Peres, Alistair Sinclair, Perla Sousi, and Alexandre Stauffer (2013). “Mobile geometric graphs: detection, coverage and percolation”. *Probab. Theory Related Fields*
- ▶ Benyuan Liu, Olivier Dousse, Philippe Nain, and Don Towsley (2012). “Dynamic coverage of mobile sensor networks”. *IEEE Transactions on Parallel and Distributed systems*

Functionals of Interest

$$f_n(t) := \sum_{\mathcal{S} \subseteq \eta_n(t)} \xi_{k,r}(\mathcal{S})$$

for $\xi_{k,r} : \text{Fin}(\mathbb{T}^d) \rightarrow [0, \infty)$ s.t.

- **Fixed Size:** $\xi_{k,r}(\mathcal{S}) = 0$ if $|\mathcal{S}| \neq k$
- **Local:** $\exists \rho > 0$ s.t. $\text{diam}(\mathcal{S}) > \rho r$ implies $\xi_{k,r}(\mathcal{S}) = 0$
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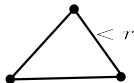
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$$\xi_{k,r}(\mathcal{S}) = 1\{\mathcal{S} \text{ is isomorphic to } \mathcal{G}\}$$

Results - I

$$\bar{f}_n(t) := \frac{f_n(t) - \mathbb{E}[f_n(t)]}{\sqrt{\text{var}[f_n(t)]}}$$

► $n \rightarrow \infty, r \rightarrow 0, \sigma \rightarrow 0$

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$$\{\bar{f}_n(t) : t \geq 0\} \xrightarrow{\text{fdd}} \left\{ \sum_{j=1}^k c_j \mathcal{U}_j(t) : t \geq 0 \right\} \text{ for some } c_j > 0.$$

Results - I

- ▶ See O, Omer Bobrowski, and Robert J. Adler (2023a). “Functional central limit theorems for local statistics of spatial birth–death processes in the thermodynamic regime”. *Ann. Appl. Probab.* for extended results.

Results - II

$$\bar{f}_n(t) := \frac{f_n(t) - \mathbb{E}[f_n(t)]}{\sqrt{\text{var}[f_n(t)]}}$$

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$\{\mathcal{V}_j^\beta(t) : t \geq 0\}$: stationary Gaussian process with covariance

$$\text{cov}[\mathcal{V}_j^\beta(t_1), \mathcal{V}_j^\beta(t_2)] = e^{-j|t_1-t_2|} \zeta_j(\beta|t_1-t_2|) \quad \text{where}$$
$$\zeta_j(t) \asymp t^{-d(j-1)/2} \quad \text{as } t \rightarrow \infty.$$

Theorem 2.

$$\{\bar{f}_n(t) : t \geq 0\} \xrightarrow{\text{fdd}} \left\{ c_1 \mathcal{U}_1(t) + \sum_{j=2}^k c_j \mathcal{V}_j^\beta(t) \right\}$$

Results - III

$$\tilde{f}_n(t) := \frac{f_n(t) - \mathbb{E}[f_n(t)]}{\sqrt{2M_n \text{var}[f_n(t)]}}, \quad M_n := \int_0^1 \mathbb{E}[\bar{f}_n(0)\bar{f}_n(t)]dt$$

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$\{W(t) : t \geq 0\}$: standard Brownian motion in $\mathbb{R}$

Theorem 3.

$$\left\{ \int_0^t \tilde{f}_n(s) ds : t \geq 0 \right\} \xrightarrow{\text{fdd}} W(t).$$

Proof Idea

► Cramer-Wold

$$\sum_{i=1}^m \omega_i f_n(t_i) = \sum_{\mathcal{S} \subseteq \eta(n,T)} \xi_{k,r}(\mathcal{S}(t_i)) \sum_{i=1}^m \omega_i \mathbb{1}\{\text{Each point in } \mathcal{S} \text{ alive at } t_i\}$$

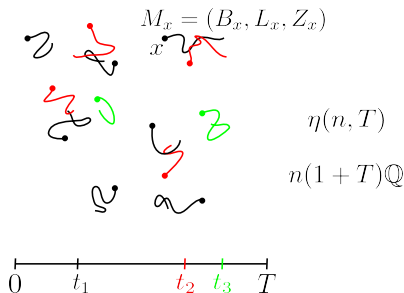
for all $0 \leq t_1 \leq \dots \leq t_m \leq T$.

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Tools

- $\sum_{i=1}^m \omega_i f_n(t_i)$ is a ‘ U -statistics’:

$$\sum_{i=1}^m \omega_i f_n(t_i) = \sum_{\mathbf{x} \in \eta_{\neq}^k} h(\mathbf{x})$$

with

$$h(\mathbf{x}) := \frac{1}{k!} \sum_{i=1}^m \omega_i \xi_{k,r}(\mathbf{x}(t_i)) \mathbb{1}\{\text{Each point in } \mathbf{x} \text{ alive at } t_i\}.$$

Gaussian Approximation on Poisson Space - Malliavin-Stein Approach

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for every continuous and piecewise cont. differentiable g
with $\mathbb{E}|g'(W)| \leq \infty$.

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- Any square integrable random variable W admits the chaotic expansion:

$$W = \mathbb{E}[W] + \sum_{q=1}^{\infty} I_q(w_q)$$

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- $\mathbb{E}[Wg(W)] = \mathbb{E}[\langle DW, -DL^{-1}W \rangle g'(W)]$

Malliavin-Stein on U -statistics

- ▶ Raphaël Lachièze-Rey and Giovanni Peccati (2013).
“Fine Gaussian fluctuations on the Poisson space, I: contractions, cumulants and geometric random graphs”. *Electron. J. Probab.*
- ▶ Matthias Reitzner and Matthias Schulte (2013).
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Corollary.

Let H_n be a U -statistic and assume $\text{var}[H_n] \rightarrow \rho^2 > 0$. Then

$$d(H_n - \mathbb{E}[H_n], \rho\mathcal{N}) \lesssim \max_{1 \leq q \leq k} \|h_q\|_{L^4(\mu^q)}^2.$$

Moment Calculations - Definitions

Definition (Intersection Pattern)

Let $\mathcal{I}_\ell$ denote a collection of natural numbers indexed by nonempty subsets of $[\ell] = \{1, \dots, \ell\}$, i.e. $\mathcal{I}_\ell = (I_J)_{J \subset [\ell], J \neq \emptyset}$. Given a sequence of sets $\mathcal{S}_1, \dots, \mathcal{S}_\ell \in \text{Fin}(\mathbb{R}^d)$, suppose that for all nonempty $J \subset [\ell]$ we have

$$\left| \left(\bigcap_{j \in J} \mathcal{S}_j \right) \cap \left(\bigcap_{j \notin J} (\mathcal{S}_j)^c \right) \right| = I_J,$$

In this case we say that $\mathcal{S}_1, \dots, \mathcal{S}_\ell$ obey the **intersection pattern** $\mathcal{I}_\ell$, and denote this by $(\mathcal{S}_1, \dots, \mathcal{S}_\ell) \in \mathcal{I}_\ell$. Set

$$|\mathcal{I}| := \sum_J I_J = |\mathcal{S}_1 \cup \dots \cup \mathcal{S}_\ell|.$$

Moment Calculations - A Combinatorial Lemma

Lemma (Generalized Mecke Equation)

Given a sequence of points $\mathbf{x} = (x_1, \dots, x_{|\mathcal{I}|})$ in $\mathbb{R}^d$, let

$$\Pi_{\mathcal{I}}(\mathbf{x}) := (\mathcal{T}_1, \dots, \mathcal{T}_{\ell})$$

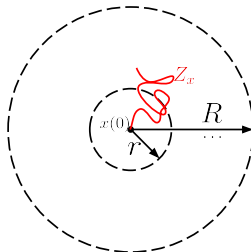
be a splitting of the tuple $\mathbf{x}$ into $(\mathcal{T}_1, \dots, \mathcal{T}_{\ell}) \in \mathcal{I}$, in an arbitrary but fixed manner. Then

$$\mathbb{E} \left[\sum_{\mathcal{S}_1 \subset \eta_n} \cdots \sum_{\mathcal{S}_{\ell} \subset \eta_n} h(\bar{\mathcal{S}}, \eta_n) \mathbb{1}\{\bar{\mathcal{S}} \in \mathcal{I}\} \right] \asymp n^{|\mathcal{I}|} \mathbb{E} [h(\Pi_{\mathcal{I}}(\mathbf{X}), \mathbf{X} \cup \eta'_n)]$$

where $\bar{\mathcal{S}} := (\mathcal{S}_1, \dots, \mathcal{S}_{\ell})$, and $\mathbf{X}$ is a tuple of $|\mathcal{I}|$ iid points in $\mathbb{R}^d$, with density q , and independent of η'_n .

Challenges due to mobility

$$A := \{\|Z_x(\Delta)\| \leq R \text{ for all } x \in \eta_n\}$$

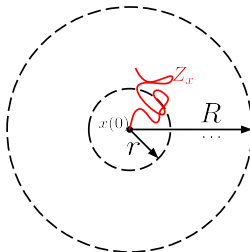


- While calculating $\text{cov}[f_n(0)f_n(\Delta)]$, choose R such that

$$\text{cov}[f_n(0)f_n(\Delta)] \approx \mathbb{P}[A]\text{cov}[f_n(0)f_n(\Delta)|A]$$

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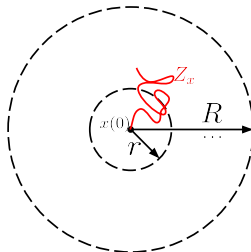
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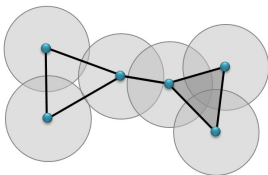
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- ▶ In the moderate regime take $R := \sigma\sqrt{\Delta r^{-\epsilon} \log r^{-1}}$.
- ▶ In the fast regime take $R := \sigma\sqrt{\Delta} \log \frac{1}{\sigma\sqrt{\Delta}}$.

Extensions

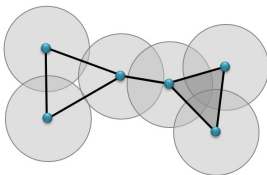
- Topological invariants of geometric simplicial (e.g. Čech, Vietoris-Rips) complexes



Omer Bobrowski and Matthew Kahle (2018).
“Topology of random geometric complexes: a survey”. *J. Appl. Comput. Topol.*

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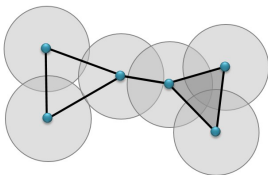
Theorem 4 (O, Bobrowski, and Adler 2023a).

If $nr^d \rightarrow \gamma \in (0, \infty)$, $\sigma = 0$ then

$$\{\bar{F}_n(t) : t \geq 0\} \xrightarrow{\text{cadlag}} \left\{ \sum_{j=1}^{\infty} c_j \mathcal{U}_j(t) : t \geq 0 \right\}.$$

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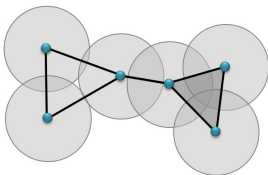


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- Other properties of the process (exceptional times etc.)
- Arbitrary closed manifolds, other dynamics, models...

Thank you.

Spatial Birth Death Processes

- ▶ Chris Preston (1975). “Spatial birth-and-death processes”. *Bull. Inst. Internat. Statist.*
- ▶ Lorenzo Bertini, Nicoletta Cancrini, and Filippo Cesi (2002). “The spectral gap for a Glauber-type dynamics in a continuous gas”. *Ann. Inst. H. Poincaré Probab. Statist.*
- ▶ M. N. M. van Lieshout (1994). “Stochastic annealing for nearest-neighbour point processes with application to object recognition”. *Adv. in Appl. Probab.*
- ▶ A. J. Baddeley and M. N. M. van Lieshout (1995). “Area-interaction point processes”. *Ann. Inst. Statist. Math.*